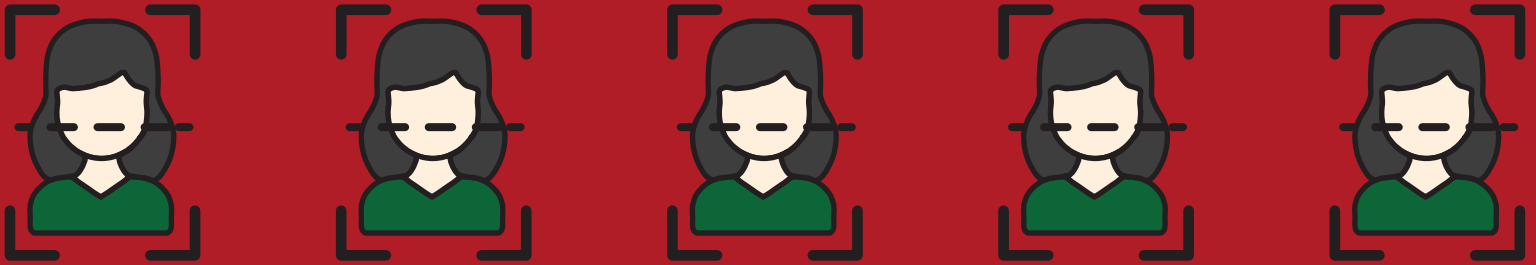


Dynamic Face Perception: The Role of Expertise in Dual Processing of Features and Configuration

Yinqi Huang¹ 



ABSTRACT

Face perception is the basis of many types of social information exchange, but there is controversy over its underlying mechanisms. Researchers have theorized two processing pathways underlying facial perception: configural processing and featural processing. Featural processing focuses on the individual features of a face, whereas configural processing focuses on the spatial relations of features. To resolve the debate on the relative contribution of the two pathways in face perception, researchers have proposed a dual processing model that the two pathways contribute to two different perceptions, detecting face-like patterns and identifying individual faces. The dual processing model is based on face perception experiments that primarily use static faces. As we mostly interact with dynamic faces in real life, the generalization of the model to dynamic faces will advance our understanding of how faces are perceived in real life. This paper proposes a refined dual processing model of dynamic face perception, in which expertise in dynamic face perception supports identifying individual faces, and it is a learned behaviour that develops with age. Specifically, facial motions account for the advantages of dynamic faces, compared to static faces. This paper highlights two intrinsic characteristics of facial motions that enable the advantages of dynamic faces in face perception. Firstly, facial motion provides facial information from various viewpoints, and thus supports the generalization of face perception to the unlearned view of faces. Secondly, distinctive motion patterns serve as a cue to the identity of the face.

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Published online on February 23, 2023

Introduction

Face perception is the basis of many types of social information exchange because faces provide visual information vital for social interaction, such as group membership, emotional states, and attention^{1,2}. Both behavioural and neurobiological evidence suggest that there is a complex mechanism underlying face perception, but there is a debate on the cognitive processes that comprise it. Researchers have theorized two processing pathways: featural processing and configural processing³⁻⁶. Featural processing focuses on the individual features of a face, whereas configural processing focuses on the spatial relations of the features. These two pathways have different development rates⁷⁻⁹ and are associated with different ERP markers^{10,11} and different brain regions¹²⁻¹⁴. Because the two pathways have distinguishable behavioural outcomes and neuroanatomical localization, the two pathways are considered separable yet intertwined.

Though researchers agree that the two pathways both contribute to face perception, they have been debating the relative contributions of the two. Some support a configuration-dominant view that face perception is dominated by the perception of the configuration

of features^{4,15}, while others argue for a feature-dominant view that isolated features can support face perception¹⁶ and are processed independently from the configuration¹⁷. Some researchers also take on a moderate view that both pathways contribute to face perception, with configural-processing playing the main role^{5,6}. To settle this debate, de Gelder theorized a dual processing model, which states that the two pathways contribute to different levels of face perception¹⁸. However, de Gelder's model is based on face perception experiments that use primarily static faces. As we mostly interact with dynamic faces in real life, generalization of the model to dynamic faces will advance our understanding of how faces are perceived in real life. Researchers have observed that dynamic faces are more identifiable than static faces, and they have proposed factors that explain this advantage¹⁹⁻²¹. Built on the dual processing model, this paper proposes a refined model in which expertise of dynamic faces supports face perception. This paper first reviews configural processing and featural processing; it then discusses behavioural and neurobiological evidence supporting the dual processing model; finally, it reviews behavioural studies that support the refined dual processing model. The refined model proposes that dynamic face perception supports face identification

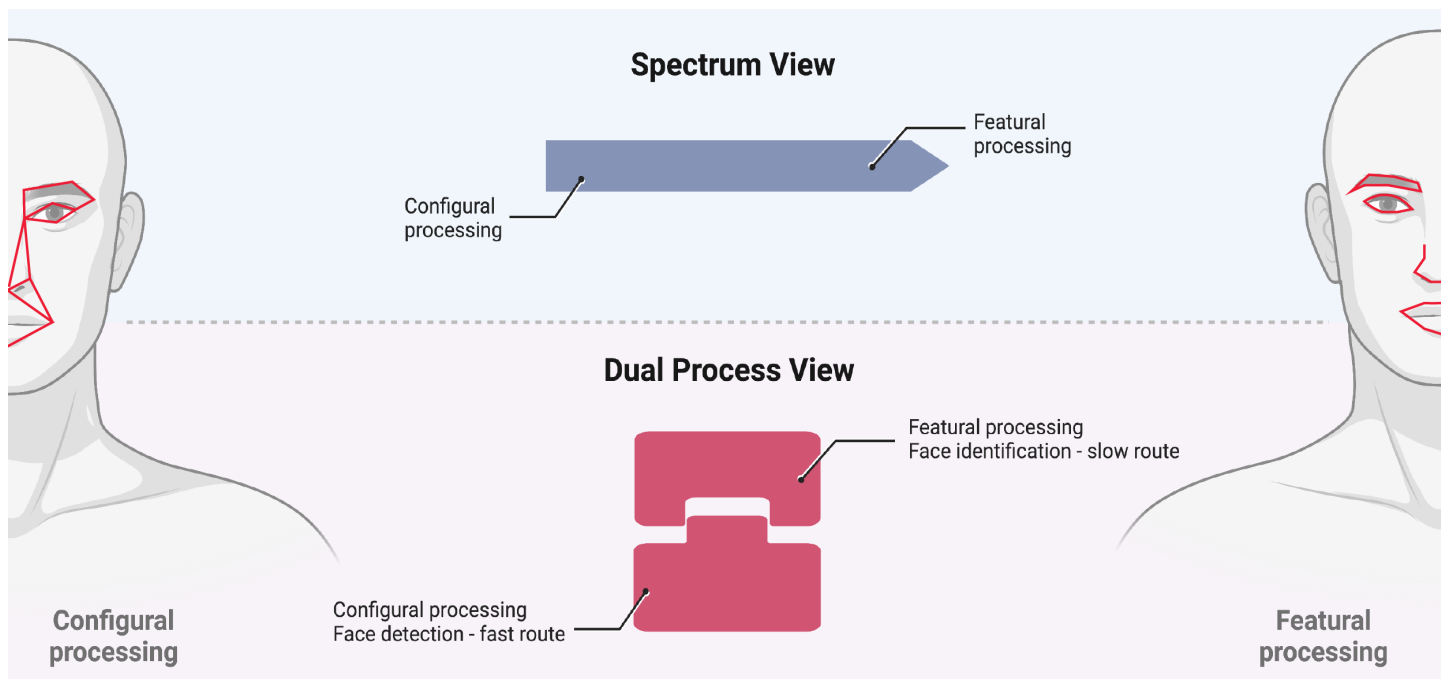


Figure 1. Configural and featural processing in the spectrum view and the dual process view. Configural processing encodes both facial features and spatial distance, whereas featural processing encodes individual facial characteristics, such as the shapes of the eyes, nose, and mouth. The spectrum view suggests either strategy is sufficient for face perception, while the dual processing model argues both are necessary for different levels of perception. The fast route, supported by configural processing, enables rapid face detection, while the slow route, supported by featural processing, enables goal-driven face identification. Graphics created with BioRender.com.

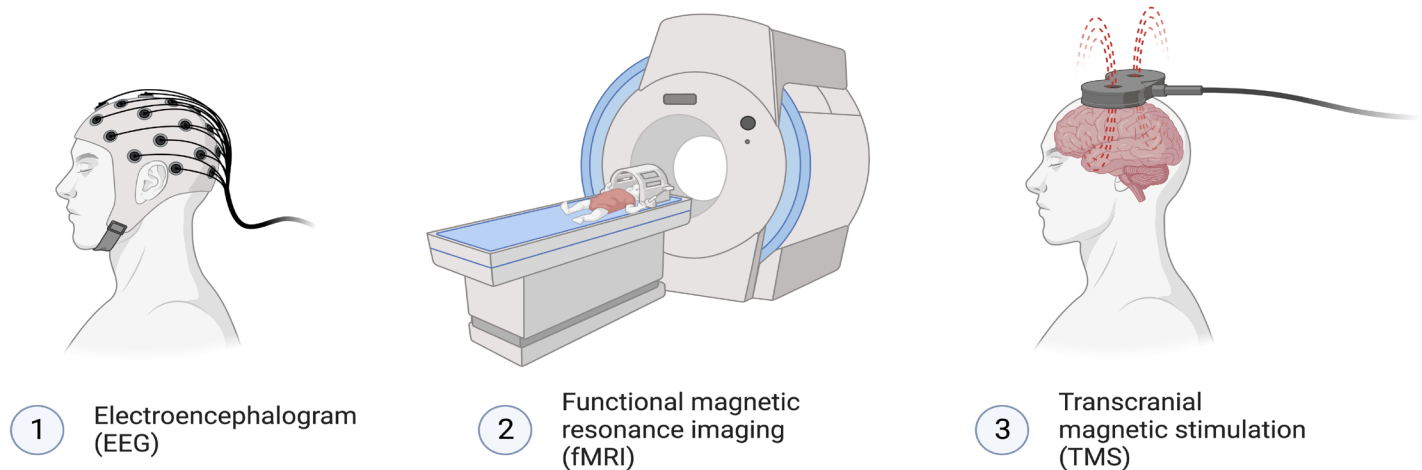


Figure 2. Temporal and spatial precision of EEG, fMRI, and TMS in studying face perception. EEG has exceptional temporal precision, allowing for the detection of fast changes in whole brain activity related to face processing. However, EEG has relatively poor spatial precision and cannot be used to pinpoint the exact brain regions. In contrast, fMRI has excellent spatial precision, which can be useful to identify the precise location related to face perception. However, fMRI has relatively poor temporal precision, making it less suitable for studying fast neural changes. TMS, on the other hand, has excellent temporal precision and can study fast neural changes and establish a causal link between stimulated regions and behavioural changes. However, TMS has relatively poor spatial precision because it may affect neighboring regions around the targeted brain region. The method chosen is dependent on the research question and specific neural process being studied in face perception. Graphics created with BioRender.com.

and it is a learned behaviour that develops with age.

Featural Processing versus Configural Processing

Featural processing in face perception refers to the theorized processing strategy where the characteristics of individual facial features, such as the shapes of the eyes, nose, and mouth are encoded, whereas configural processing refers to the strategy where the features along with spatial distance are encoded. Researchers managed to discern three types of configural processing: first-order processing, second-order processing, and holistic processing²². First-order refers to the relative position of features that generates a face-like stimulus (e.g., two eyes above a mouth), whereas second-order refers to the spatial distances among features, contributing to the individualization of faces. Holistic processing refers to the process that integrates features into a gestalt. This paper refers to all three types of processing as configural processing to avoid verbiage, because the discussion focuses on the higher level of categorization on configural processing and featural processing.

Though some might argue that it is impossible

to separate featural processing from configural processing because configural processing always involves some degree of featural representation, there are techniques to prioritize the employment of one of the processing pathways. For example, Lobmaier and his colleagues designed a blurred face stimulus where the featural configuration was preserved, but the blurred features were unidentifiable²³. The assumption was that the study participants would prioritize the processing of configural information because featural information is not informative. Similarly, he designed a scrambled face stimulus where the facial features were intact, but the configuration of features was scrambled²³, thus participants would prioritize featural processing. Together with a clinical face-blind condition, congenital prosopagnosia, the researchers were able to partition the two pathways behaviourally. Congenital prosopagnosia refers to the impairment of face perception that is evident from early childhood in the absence of manifest brain injury. Lobmaier found that prosopagnosic patients performed worse in identifying familiar faces when they were blurred than scrambled²³, suggesting the deficit happened when the patients employed configural processing while featural processing was discouraged by blurred facial features. Another technique is to crop a face stimulus

horizontally and misalign the top- and bottom-half of the face, thus facial features were preserved, but the featural configuration was distorted. Compared to controls, prosopagnosic patients paradoxically performed similarly on identifying aligned and misaligned faces²⁴, suggesting that with intact features, they employed featural processing which did not hinder face perception. Together, the studies demonstrate that prosopagnosic patients process featural and configural information independently, and that researchers can design the experiment in a way that participants selectively employ one of the two processing pathways.

Not only can the two processing pathways be partitioned behaviourally, but neurobiological evidence supports the separation of the two pathways. Researchers have identified different Event-Related Potentials (ERPs) associated with configural processing and featural processing, respectively^{10,11}. Mercure reported that P1 ERP components were more sensitive to the changes in the configuration of face stimuli, whereas P2 was more sensitive to featural changes¹⁰. P300 was more sensitive to featural changes, mediating through attention¹⁰. This evidence shows that featural and configural changes are processed at different times and are associated with different neural coding. Additionally, using functional magnetic resonance imaging (fMRI), researchers have identified brain regions that correlate with configural processing and featural processing, respectively^{12,13}. The left fusiform gyrus, the left parietal lobe, and the left lingual gyrus activated when participants were processing configural information, whereas the middle temporal gyrus activated when processing featural information¹³. ERPs and fMRI data can reveal correlation relationships, whereas brain stimulation techniques such as Transcranial Magnetic Stimulation (TMS) provide strong evidence because they establish a causal link through stimulating brain

nerve cells with a series of short magnetic pulses, linking the stimulated brain region with behavioural changes. A study using TMS demonstrated a double dissociation of brain regions for configural processing and featural processing¹⁴. The researchers applied TMS during a face identification task, and they found that applying TMS on the left middle frontal gyrus only interfered with featural processing, whereas applying TMS on the right inferior frontal gyrus interfered with configural processing¹⁴. Together with the observation that the two processing pathways mature at different rates in children^{7-9,22}, it is undeniable that the two processing pathways have separate neural underpinnings and behavioural outcomes.

The Dual Process Model of Face Perception

Though researchers agree that both processing pathways contribute to face perception, they have been debating the relative contributions of the two. Their views fall on a spectrum. On one end, some researchers support the configuration-dominant view that face perception is dominated by the perception of the configuration of features^{4,15}; on the other, researchers argue for the feature-dominant view that isolated features can support face perception¹⁶ and are processed independently from the configuration¹⁷. Some researchers also take a moderate position that both pathways contribute to the same level of face perception, with configural-processing playing the main role^{5,6}. To propose an alternative to this debate, de Gelder disregarded the spectrum framing entirely and proposed a dual processing model in which both processing pathways are necessary but contribute to different levels of face perception¹⁸.

The dual processing model states that face perception comprises two systems: face detection

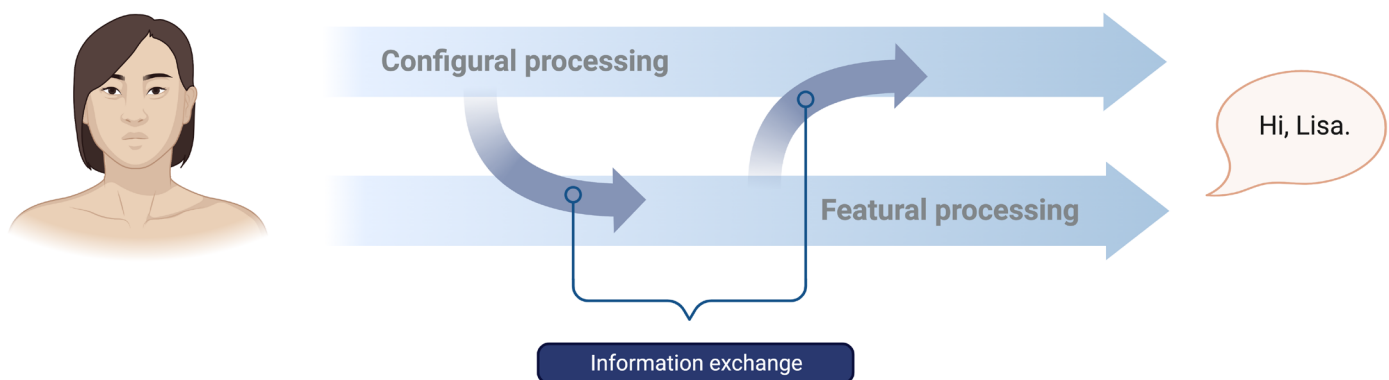


Figure 3. Information exchange during dual processing of face perception. The dual processing model suggests parallel processing, but de Gelder emphasizes dynamic information exchange between configural and featural representations. Graphics created with BioRender.com.

and face identification¹⁸. Face detection refers to the ability of detecting face-like patterns, whereas face identification refers to the ability of identifying individual faces. The separation of face detection and face identification explains the paradoxical observation that prosopagnosic patients who had normal face detection could not identify familiar faces¹⁸, as the model proposes that the two systems are functionally dissociable. The separation of the two systems is also supported by ERP evidence that facial structures were encoded before face identification²⁵. Furthermore, the model proposes that face detection is supported by a fast route and face identification is supported by a slow route¹⁸. The characteristics of configural processing and featural processing also support the fast and slow route proposals. Face detection is a rapid process sensitive to external stimuli such as colour and shape, and it only encodes coarse-grained representations of the stimuli, such as the configuration of face-like patterns. On the contrary, the slow route contributes to face identification, or the ability of identifying individual faces. It is a voluntarily goal-driven process and is supported by fine-grained representations, such as featural details of a face.

The above interpretation of the dual processing model implies that configural processing and featural processing are parallel. However, de Gelder instead emphasized that this is a dynamic model, and thus the information of featural representation and configural representation are exchanged as output between the face detection system and face identification system¹⁸. Most interestingly, de Gelder proposed that face detection requires limited stimulus exposure, whereas face identification depends on extensive exposure and shares processing resources with object recognition. This is supported by the evidence that newborns who had not seen a face tracked face-like pattern more than scrambled configuration²⁶, suggesting that detection of face-like patterns is an innate response. On the other hand, the accuracy of identifying familiar upright faces improves linearly from 5 to 18 years old²⁷. In addition, the improvement increases more rapidly than identifying inverted faces, suggesting that identifying individual faces is constantly developing with age, likely due to more exposure to upright faces and improved memory²⁷.

A Refined Model of Dynamic Face Perception

Built on the dual processing model and de Gelder's proposal on face expertise¹⁸, this paper proposes a refined model in which face identification is determined by the expertise of dynamic faces, demonstrated by behavioural evidence below. Though de Gelder designed a comprehensive model, his model is based on face perception experiments that used primarily static faces. As we mostly interact with dynamic faces

in real life, generalizing his model to dynamic faces will advance our understanding of how faces are perceived in real life.

A piece of direct evidence of the difference between static face perception and dynamic face perception is that a prosopagnosic patient, who had impairment in identifying familiar static faces, was able to identify dynamic faces¹⁹, suggesting that characteristics unique to dynamic faces also supports face identification, aside of static facial features and featural configuration. Moreover, unfamiliar faces learned in motion were correctly identified more than static faces²⁰, suggesting that dynamic faces are more identifiable than static faces.

An interesting interpretation of the underlying mechanism of facial motions' advantage is that dynamic faces are not encoded better because of intrinsic characteristics of motions, but rather social animals pay more attention to moving faces, making them more memorable²⁸. However, this paper summarizes two consensus characteristics of facial motions that enable this advantage. O'Toole and her colleagues proposed that changeable motion-based information in dynamic faces provides rich communication signals supplementary to static faces, which facilitate face perception under non-optimal viewing conditions²⁹. This paper further develops the proposal by arguing that the distinctive characteristics of facial motions account for the advantages of dynamic faces in face perception.

Firstly, facial motions provide facial information from various viewpoints. Identification of static faces demonstrates dependency on the learned view of face stimuli³⁰. Familiarity of the learned view does not support identification of the unlearned view of the same face, whereas head rotations support generalization to both upright and inverted orientation³⁰, suggesting that three-dimensional facial information reduces viewpoint dependence. Secondly, motion-based facial expressions facilitate face identification. For example, Lander and her colleagues found that distinctive motion patterns, such as smiling and talking, serve as an additional cue to the identity of the face³¹. Familiarity of dynamic faces correlates with the distinctiveness of facial motions³², suggesting that faces moving in a distinctive manner benefit the most from being seen in motion. Another piece of evidence is that the parents of monozygotic twins, or identical twins, were able to distinguish between their own twins' who they interacted with in real life³³. Given that facial features and configurations are identical for the twins, this finding suggests that the familiarity of the twins' facial motions contributes to the successful identification of their faces.

Based on the refined model that expertise of dynamic

faces accounts for the advantages of dynamic faces, we should expect that the dynamic face perception would be more successful when the observer is more familiar with the facial motion. This prediction is supported by the finding that infants gradually develop a looking preference for dynamic faces over dynamic abstract stimuli with age³⁴. This preference appears later than the preference for static faces, indicating that the perception of dynamic faces is a learned behaviour.

Discussion

In conclusion, built on the dual processing model that integrates featural processing and configural processing, this paper proposes a refined model of dynamic face perception by drawing evidence from the functional roles of facial motions and the gradual acquisition of preference for dynamic faces in infants. As we are more likely to encounter moving faces in real life, for future studies, it would be interesting to investigate whether the refined expertise model could predict dynamic face perception in both face detection and face identification task. An interesting direction would be investigating whether the expertise of the dynamic faces of similar-looking individuals (e.g., the familiarity of identical twins' continuous facial movements) contributes to the identification of their faces, as identical twins' parents are slower at distinguishing between unfamiliar twins than between their own twins' who they interact with in real life, when looking at photographs after learning³³. With the recent advancement in photo-realistic morphing techniques and synthetic facial animations, dynamic face stimuli are more controllable as the new techniques allow for sophisticated manipulation of facial characteristics and motions. Therefore, adopting dynamic face stimuli in face perception experiments will be a novel and promising direction³⁵.

Moreover, generalizing the dual processing model to dynamic faces will provide a stepping stone to constructing computational models that capture the cognitive process underlying dynamic face perception. Based on David Marr's proposal on connecting computational, the algorithmic, and the implementational level of human visual processing³⁶, if we understand how featural and configural information of dynamic faces are processed, we can construct computational models of the cognitive process of face perception in the real world situations. This can be done through, for example, connecting behavioural and neural data of dynamic face perception by capturing non-observable variables using methods such as Bayesian inference³⁷.

Separately, modelling the cognitive processing strategies on how humans perceive and distinguish similar-looking individuals will allow for the optimization

of face recognition algorithms which label images of faces. Similar to the human perception of identical twins, human observers who interact with dynamic faces still outperform computer classifiers that are trained on static faces in identifying identical twins' faces^{38,39}. Thus, uncovering the processing strategies may allow for the optimization of face recognition algorithms as modern artificial neural networks' architectures frequently get inspiration from animal brain models³⁹. Notable examples include Spiking Neural Networks (SNN) inspired by synaptic transmission⁴¹, Convolutional Neural Networks (CNN) inspired by visual processing system⁴², and Recurrent Neural Networks (RNN) inspired by the backpropagation of errors in the brain^{43,44}.

COMPETING INTERESTS

No competing interests declared.

ACKNOWLEDGEMENTS

The author would like to express gratitude to the esteemed reviewers, whose thoughtful and constructive feedback greatly enhanced the quality of this manuscript. Appreciation is also extended to the dedicated editorial teams for their invaluable assistance throughout the submission and review process.

Received: September 8, 2021

Accepted: January 12, 2022

Published online: February 23, 2023

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